COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 5 : SLO – 1 | Memory Locations and Addresses
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Section A – Match the Following (Answers)

• 1. Word Length → D. Number of bits grouped and accessed together
• 2. Byte → C. A group of 8 bits
• 3. Main Memory → A. Collection of storage locations with unique identifiers
• 4. Address → E. Unique identifier for a word in memory
• 5. Address Space → B. Minimum number of bits needed to represent all memory locations

Section B – Fill in the Blanks

• The term BYTE refers to a group of 8 bits.
• A word of 16 bits is commonly referred to as a HALF word.
• The total number of uniquely identifiable locations in memory is called the ADDRESS SPACE.
• Each storage location in memory has a unique identifier called an ADDRESS.
• To access 32 MB of memory byte-wise, we need 25 bits for addressing (32 MB = 2^25 bytes).

Section C – Activity Task

• Task 1a: 64 MB memory, word = 4 bytes → number of words = 64M/4 = 16M = 2^24 → 24 bits are needed to address a word.
• Task 1b: 1 GB memory, word = 8 bytes → number of words = 2^30 / 2^3 = 2^27 → 27 bits are required to address each word.
• Task 2 – Memory layout (described): a column of words from address 0 (first word) to 2^k – 1 (last word); each row is one n-bit word; the i-th word sits at address i; addresses label the rows, the n-bit contents are the data.

Section D – True or False

• A byte is made up of 16 bits. — FALSE (8 bits)
• Memory addresses are expressed using unsigned binary numbers. — TRUE
• The main memory stores data in single bits only. — FALSE
• The term 'address space' refers to the number of address lines available. — FALSE
• In computer hardware, names are used to identify memory locations. — FALSE (binary addresses are used)